

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	18 July 2026
Team ID	
Project Name	HouseHunt – Premium House Rental Marketplace
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Registration through Gmail Registration through LinkedIn
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Property Management	Add / edit / delete a property listing Publish or save a listing as draft Upload listing images and amenities Admin approval required before a listing goes live
FR-4	Booking & Negotiation	Submit a property tour / booking request Confirm or cancel a booking Submit a rent negotiation offer Counter-offer or accept a negotiation
FR-5	Wallet & Payments	View wallet / transaction ledger Submit an owner withdrawal request Automatic GST and platform commission calculation
FR-6	Chat & Complaint Management	Real-time chat between renter and owner Register a complaint Track and resolve a complaint (Admin)

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Responsive interface (Tailwind CSS, Material UI, Bootstrap) with role-based dashboards for renter, owner and admin across desktop and mobile.
NFR-2	Security	JWT-based authentication, bcryptjs password hashing, and role-guard middleware enforce access control on every protected route.
NFR-3	Reliability	Mongoose schema validation and centralized Express error-handling middleware maintain data integrity and consistent API responses.
NFR-4	Performance	Vite-based fast frontend builds, an Axios API layer and indexed MongoDB queries keep search and dashboard interactions responsive.
NFR-5	Availability	Deployed on Vercel with the client as a static site and the Express server as a serverless API, backed by a managed MongoDB cluster.
NFR-6	Scalability	A three-tier architecture with modular controllers/services separates presentation, application and data layers, allowing new modules (payments, mobile apps) to be added without reworking the core platform.